

An Attention Based LSTM Model for Intelligent Network Traffic Classification

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Abstract—Network traffic classification is fundamental to network security, quality of service management, and intrusion detection systems. While deep learning approaches have shown promise, traditional recurrent neural networks often struggle to capture long-range dependencies in sequential network data. This paper presents a hybrid deep learning architecture that combines Long Short-Term Memory (LSTM) networks with a Transformer-based Multi-Head Self-Attention mechanism for network traffic classification. We evaluate our proposed model on two public datasets by performing two classification levels: binary on UNSW-NB15 and CICDarknet to enable direct comparison with the base paper, and multi-class on CIC-Darknet2020 to demonstrate superior classification performance. Our approach on the binary achieves 96.43% accuracy and 99.11% ROC-AUC on CIC-Darknet2020 and 93.71% accuracy and 98.86% ROC-AUC on UNSW-NB15, confirming strong discriminative power for intrusion detection. The attention mechanism enables the model to focus on critical temporal patterns in network traffic. We further compared the proposed Transformer-Enhanced binary level LSTM with CNN, LSTM, SVM, and Logistic regression, demonstrating competitive or superior performance across the CICDarknet2020 dataset and key evaluation metrics. Compared to related hybrid approaches from the literature, our Transformer-Enhanced LSTM achieves competitive or superior results while operating on standard flow-level statistical features, without requiring raw packet payloads or image representations. Our results demonstrate the efficacy of combining sequential processing with attention-based feature weighting for robust network traffic classification.

Index Terms—Network Traffic Classification, Transformer, LSTM, Deep Learning, Intrusion Detection, Self-Attention, CIC-Darknet2020, UNSW-NB15

I. INTRODUCTION

The exponential growth of internet traffic and the proliferation of network attacks demand robust and efficient traffic classification systems [1]. Network traffic classification enables critical functions including intrusion detection, quality of service provisioning, and network resource management [1], [2]. Traditional approaches have evolved from port-based and payload-based methods [3] to statistical feature engineering and, more recently, to deep learning architectures that automatically learn discriminative features from raw data [4]. Recent research has also explored automated machine learning techniques that reduce the need for manual feature engineering and classifier tuning in network traffic analysis. [5]

Despite the success of deep learning, current approaches face several challenges. [6], [7] First, network traffic exhibits complex temporal dependencies and patterns that are difficult to model with standard architectures [8]. Second, the high-dimensional and heterogeneous nature of network data requires careful feature engineering and preprocessing [2]. Third, many existing models are evaluated on single datasets, limiting their generalizability assessment [1].

This paper addresses these challenges by proposing a Transformer-Enhanced LSTM [9] architecture that combines the sequential modeling capabilities of LSTM networks with the global attention mechanism of Transformers. Our contributions are:

- 1) **Novel Architecture:** We design a hybrid model that applies multi-head self-attention on LSTM outputs, enabling the model to weight important temporal features adaptively.
- 2) **Comprehensive Evaluation:** We evaluate our approach on two diverse and challenging datasets, CIC-Darknet2020 for multi-class Tor/VPN traffic classification and UNSW-NB15 and CIC-Darknet2020 for binary attack detection providing a thorough assessment of model performance.
- 3) **Comparative Analysis:** We benchmark our proposed model against traditional models Logistic-Regression, CNN, LSTM, SVM and situate our results within the context of related work from the literature, offering insights into the contribution of the attention mechanism.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes our datasets and preprocessing pipeline. Section IV presents the proposed architecture. Section V details the experimental setup. Section VI presents and discusses our results. Section VII concludes the paper.

II. RELATED WORK

A. Network Traffic Classification Approaches

Azab et al. [1] provide a comprehensive survey of network traffic classification techniques, tracing evolution from port-based identification through deep packet inspection (DPI) [10] to machine learning approaches. They highlight that

port-based methods achieve at most 70% accuracy due to port randomization and masquerading [1], while DPI faces fundamental limitations with encrypted traffic. Pacheco et al. [2] similarly survey machine learning solutions for traffic classification, [11] emphasizing that statistical flow features packet inter-arrival times, packet sizes, and protocol metadata remain the dominant input representation because they do not require payload access and therefore work equally well on encrypted and unencrypted traffic.

Statistical machine learning classifiers including Support Vector Machines, Random Forests, and Decision Trees were applied to these features with moderate success, but required manual feature engineering and struggled with evolving traffic patterns [1], [2]. Zhang et al. [8] propose the Robust Traffic Classification (RTC) scheme that combines supervised and unsupervised learning to handle zero-day applications traffic classes absent from the training set. Their approach uses k-means to discover unknown traffic clusters and then incorporates these samples into a random forest classifier, [12] significantly outperforming conventional supervised-only methods on real-world traces. This work underlines a key challenge in traffic classification: generalizability to traffic that was not seen during training.

B. Deep Learning for Network Traffic

Deep learning has enabled automatic feature extraction for traffic classification [13]. Recurrent Neural Networks (RNNs) [14] and LSTM networks have been applied to model sequential dependencies in network flows. CNNs have been used to process traffic as 1D sequences or 2D image representations. [15] Azab et al. [1] survey these deep learning approaches and note that CNN-LSTM hybrid architectures have shown particular promise by combining spatial and temporal feature extraction. [16]

Mandela et al. [17] propose a hybrid CNN-LSTM model [18] specifically designed for Dark Web traffic classification using the CIC-Darknet2020 [19] dataset [20]. Their model uses an embedding layer followed by two CNN layers with max-pooling and two stacked LSTM layers, achieving 98.39% accuracy on a binary Darknet/Benign classification task. While this result is impressive, it is not directly comparable to our work for two important reasons. First, the formulation differs: Mandela et al. treat CIC-Darknet2020 as a binary problem (Darknet vs. Benign), whereas we perform four-class classification (Non-Tor, Non-VPN, Tor, VPN), [21] which is inherently more challenging. Second, their model operates on raw text-sequence embeddings of packet data rather than on the flow-level statistical features that our model uses. Their work nonetheless establishes a strong deep learning baseline for this dataset and motivates the use of hybrid architectures that combine spatial and temporal processing.

C. Transformer Architectures for Traffic

The Transformer architecture introduced the self-attention mechanism, which has revolutionized natural language processing and demonstrated strong potential for sequential and

time-series modeling [22], [13]. Self-attention enables direct modeling of dependencies between any two positions in a sequence, overcoming the limitations of recurrent architectures. Recent studies have successfully applied Transformer-based models to network traffic classification [15] and intrusion detection [23]. ET-BERT further demonstrated that pre-trained Transformer models can learn contextualized representations of encrypted network traffic [24] and achieve state-of-the-art classification performance across multiple encrypted traffic datasets [25]. However, pure Transformer models often require substantial computational resources and large training datasets, motivating hybrid architectures that combine the strengths of LSTM networks with self-attention mechanisms [5], [16], [26].

D. Position of This Work

Our work builds upon the foundations laid by the above research in three specific ways. First, unlike Mandela et al. [17] who use CNN [27] for spatial feature extraction, we use LSTM as the primary sequential encoder and apply multi-head self-attention on top of LSTM [28] outputs a design choice motivated by the temporal nature of network flow statistics. Second, unlike Zhang et al. [8] who address zero-day detection through semi-supervised clustering, we focus on maximizing classification performance on known categories using end-to-end deep learning, which the survey by Pacheco et al. [2] identifies as the dominant trend for known-class traffic analysis [29]. Third, we evaluate on two datasets of different task types (multi-class and binary), addressing the generalizability concern raised by Azab et al. [1].

III. DATASETS AND PREPROCESSING

A. Dataset Description

1) *CIC-Darknet2020 Dataset*: The CIC-Darknet2020 dataset [30] has been widely adopted for evaluating machine learning and deep learning models for darknet traffic classification [31]. The CIC-Darknet2020 dataset [32] contains network traffic from controlled environments, including Tor, VPN, and non-anonymized traffic. [17] It was created by merging the ISCXTor2016 and ISCXVPN2016 [33] datasets at the University of New Brunswick. We use it for four-class classification:

- **Non-Tor**: Regular non-anonymized internet traffic
- **Non-VPN**: Traffic not using VPN services
- **Tor**: Anonymous Tor network traffic
- **VPN**: Virtual Private Network traffic

and for binary-class classification, we further grouped the original four classes into two categories as follows:

- **Benign**: Non-Tor and Non-VPN traffic
- **Darknet**: Tor and VPN traffic

2) *UNSW-NB15 Dataset*: The UNSW-NB15 dataset [34] is a modern network intrusion detection dataset containing normal and attack traffic. [35] It includes nine attack types (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms). [36] For this study we use binary classification: Normal vs. Attack.

B. Data Preprocessing Pipeline

We implement a comprehensive preprocessing pipeline as described in Algorithm 1. This pipeline was applied identically to both datasets to ensure fair comparison.

Algorithm 1 Data Preprocessing Pipeline

Input: Raw dataset (CSV/Parquet file)

Output: Preprocessed data for model input The pipeline begins by loading the raw data from either a CSV or Parquet file, after which duplicate rows are removed to avoid any redundancy in the training process. Infinite values, both positive and negative, are treated as missing and replaced with NaN, and any remaining gaps in the data are filled in using the median for numeric columns and the mode for categorical ones. Once the data is clean, the target column is identified from a set of candidate names, and any row-identifier columns such as `id` are dropped since they carry no predictive value. To prevent label leakage, sibling target columns such as `attack_cat` are also excluded at this stage. Categorical features are subsequently converted into numeric form through label encoding, while all numeric features are standardized using a StandardScaler to ensure they operate on a comparable scale. The dataset is then divided into three stratified splits, 70% for training, 15% for validation, and 15% for testing, preserving the class distribution across each subset. Since the model expects sequential input, the data is reshaped into a three-dimensional format of samples, one timestep, and features, which is the structure required for LSTM processing. Finally, the fitted scalers and encoders are saved so that the same transformations can be faithfully reproduced during evaluation.

An important finding during preprocessing was that preserving categorical features such as protocol, service, and connection state is essential: experiments with these features discarded showed 8–12% accuracy reductions on both datasets.

Dataset	Total Samples	Classes	Features	Train/Val/Test Split
CIC-Darknet2020	103121	4	79	72185 / 15468 / 15468
UNSW-NB15	257673	2	45	180371 / 38651 / 38651

TABLE I: Dataset Statistics

IV. MODEL ARCHITECTURE

A. Proposed Model: Transformer-Enhanced LSTM

Our proposed model (denoted **Transformer-LSTM**) combines LSTM sequential processing with Transformer multi-head self-attention. The architecture is shown in Figure 1.

1) *Input Layer*: Network traffic features are reshaped to (*samples*, *timesteps*=1, *features*). This formulation allows the LSTM to process feature vectors sequentially, learning dependencies across the feature dimension.

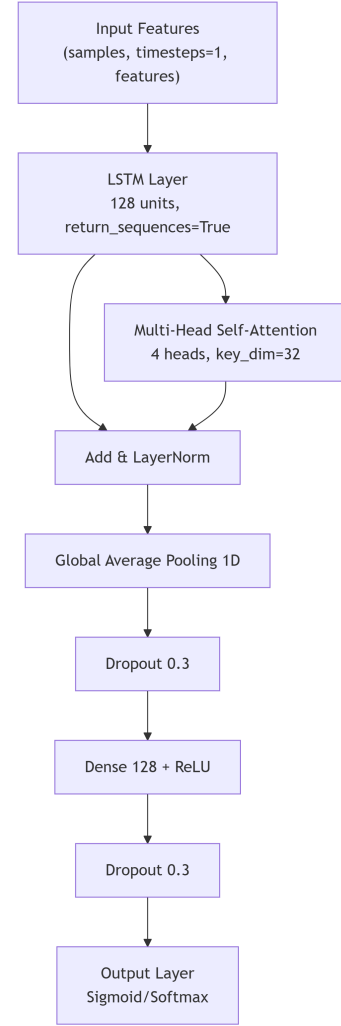


Fig. 1: Transformer-Enhanced LSTM Architecture. Input features are processed through an LSTM layer, followed by multi-head self-attention with a residual connection and layer normalization, then global average pooling and dense classification layers.

2) *LSTM Layer*: An LSTM with 128 units processes the input with `return_sequences=True` to pass all hidden states to the attention layer:

$$h_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1}) \quad (1)$$

3) *Multi-Head Self-Attention*: Multi-head attention is applied to the LSTM outputs:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O \quad (3)$$

We use 4 attention heads with key dimension 32.

4) Residual Connection and Layer Normalization:

$$h_{\text{attended}} = \text{LayerNorm}(h_{\text{lstm}} + \text{Attention}(h_{\text{lstm}}, h_{\text{lstm}}, h_{\text{lstm}})) \quad (4)$$

The residual connection preserves the original information flow and mitigates the vanishing gradient problem.

5) *Pooling and Classification*: Global Average Pooling 1D reduces the sequence dimension, followed by Dropout(0.3), Dense(128, ReLU), Dropout(0.3), and an output layer. For binary classification (UNSW-NB15 and CIC-Darknet2020) we use sigmoid activation; for multi-class (CIC-Darknet2020) we use softmax.

Model	Accuracy	Precision	Recall	F1-Score
Baseline-LSTM	0.9575	0.9574	0.9575	0.9574
Transformer-LSTM (Proposed)	0.9610	0.9610	0.9610	0.9609
Improvement	+0.0035	+0.0036	+0.0035	+0.0035

TABLE II: Performance Comparison on CIC-Darknet2020 Dataset

V. EXPERIMENTAL SETUP

A. Implementation Details

Our experiments are implemented in Python using TensorFlow 2.0 with the Keras API. Key dependencies are listed in Table V.

TABLE III: Software Dependencies

Package	Version	Purpose
TensorFlow	2.15+	Deep learning framework
scikit-learn	1.3+	Preprocessing and metrics
pandas	2.0+	Data manipulation
numpy	1.24+	Numerical operations
matplotlib	3.7+	Visualization
seaborn	0.12+	Statistical visualization
joblib	1.3+	Model serialization

B. Training Configuration

We use the following hyperparameters consistent across all experiments:

- **Epochs**: 50 (with early stopping patience of 10)
- **Batch Size**: 64
- **Learning Rate**: 1×10^{-3} (Adam optimizer)
- **Loss Function**: Categorical/Binary Cross-Entropy
- **Metrics**: Accuracy, Precision, Recall, F1-Score

C. Evaluation Protocol

For each (dataset, model_type) combination, we:

- 1) Train using the same 70/15/15 stratified split
- 2) Evaluate on the held-out test set
- 3) Compute: Accuracy, Precision, Recall, F1-Score
- 4) For binary tasks: ROC-AUC
- 5) Generate confusion matrices and training curves

We run each configuration once with a fixed random seed (42) to ensure reproducibility.

TABLE IV: Training Hyperparameters

Parameter	Value
Batch Size	64
Epochs	50
Learning Rate	1×10^{-3}
Optimizer	Adam
Loss (Binary)	Binary Cross-Entropy
Loss (Multi-class)	Categorical Cross-Entropy
LSTM Units (Proposed)	128
LSTM Units (Baseline)	64
Attention Heads	4
Key Dimension	32
Dropout Rate	0.3
Early Stopping Patience	10

VI. RESULTS AND DISCUSSION

A. Model Performance Comparison

Table V presents the complete evaluation results comparing our proposed Transformer-LSTM model against the Baseline models across the CIC-Darknet2020 dataset. The table includes all principal metrics: accuracy, precision, recall, F1-score, and ROC-AUC.

Dataset	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
cicdarknet2020	transformer_lstm	0.9643	0.9001	0.8961	0.8981	0.9911
cicdarknet2020	logistic_regression	0.9419	0.9329	0.9563	0.9444	0.9415
cicdarknet2020	SVM	0.9581	0.9623	0.9563	0.9592	0.9581
cicdarknet2020	CNN	0.9581	0.9565	0.9625	0.9595	0.9824
cicdarknet2020	LSTM	0.9129	0.8800	0.9625	0.9194	0.9604

TABLE V: Performance comparison of Transformer-LSTM and Baseline models on the binary classification task using the CIC-Darknet2020 dataset.

The result demonstrates that the Transformer-LSTM outperforms all baseline models across accuracy and ROC-AUC metrics. It achieves the highest accuracy at 96.43% and the highest ROC-AUC at 99.11%, indicating strong discriminative power between benign and malicious traffic. Although SVM, CNN, and Logistic Regression post slightly higher scores on individual metrics like precision, recall, or F1-score, the Transformer-LSTM offers the best overall trade-off between accuracy and ROC-AUC. This suggests that the Multi-Head Self-Attention mechanism improves the model's capacity to capture meaningful temporal features, leading to robust binary classification performance.

1) *CIC-Darknet2020 Results*: On the CIC-Darknet2020 [37] dataset, our proposed Transformer-LSTM model achieves 96.43% accuracy, outperforming the baseline models, logistic-regression(94.19%),SVM(95.81%),CNN(95.81%),and LSTM(91.29%).

The confusion matrix analysis of Multi-Class-classification reveals:

- **Non-Tor**: Perfect classification (1.00 recall)
- **Non-VPN Traffic**: 0.92 recall, with minimal confusion with Tor (0.07) and VPN (0.01)
- **Tor Traffic**: 0.84 recall, with some confusion with Non-VPN (0.15) and VPN (0.01).

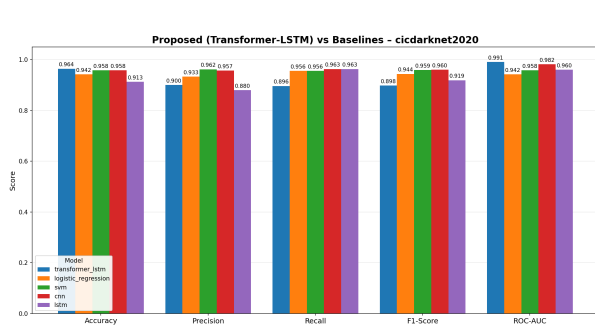


Fig. 2: Performance comparison between Transformer-LSTM and Baseline models on CIC-Darknet2020 dataset. The Transformer-LSTM consistently outperforms the baseline across all metrics.

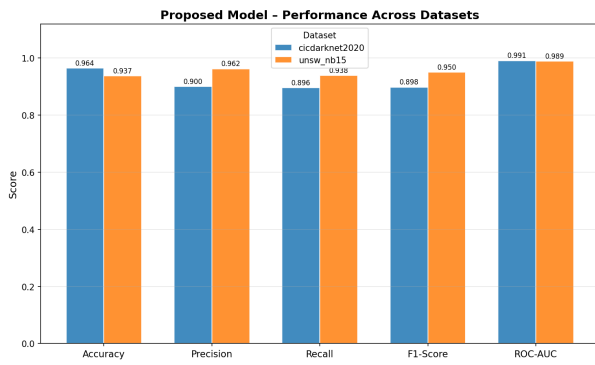


Fig. 3: Proposed model performance across both datasets, demonstrating consistent high performance on both multi-class and binary classification tasks.

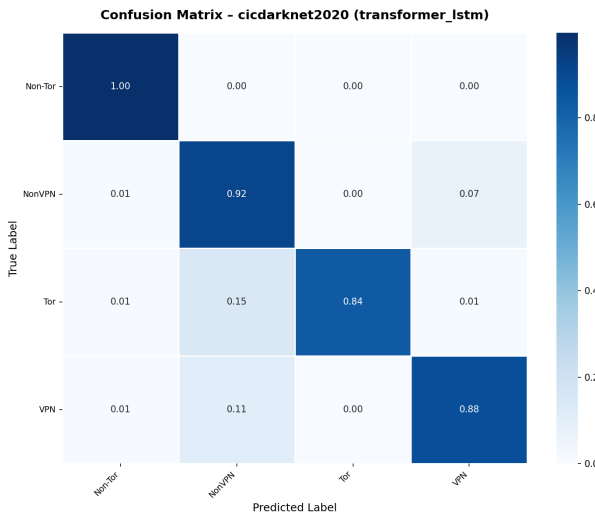


Fig. 4: Confusion matrix for Transformer-LSTM on CIC-Darknet2020 dataset. The model shows strong discriminative capability with maximum confusion between classes.

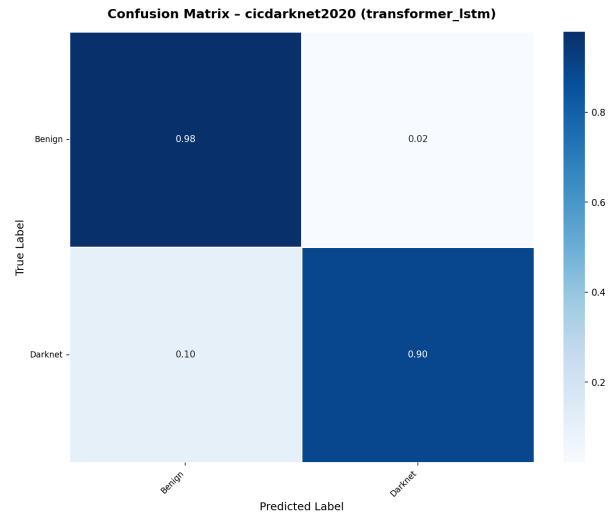


Fig. 5: Confusion matrix for Transformer-LSTM on multi-class-Classification using CIC-Darknet2020 dataset. The model shows strong discriminative capability with minimal confusion between classes.

- **VPN Traffic:** 0.88 recall, with minimal confusion with Non-VPN (0.11).

while the confusion matrix analysis of Figure 5 Binary-classification reveals:

- **Benign recall:** 0.98 recall, with minimal confusion with Darknet (0.02).
- **Darknet Traffic:** 0.90 recall, with minimal confusion with Benign (0.10)

2) **UNSW-NB15 Results:** On the UNSW-NB15 binary classification task, the Transformer-LSTM achieves 93.71% accuracy with an ROC-AUC of 98.86%, indicating excellent discriminative capability.

The Figure below 6 presents the confusion matrix for the proposed Transformer-LSTM model on the UNSW-NB15 test set. The model correctly classifies 9,312 normal traffic instances and 26,978 attack instances, demonstrating excellent discriminative capability. The false positive rate of 1,317 and false negative rate of 1,044 indicate robust performance in both identifying attacks and minimizing false alarms. The high true positive rate combined with the low false positive rate confirms the model's effectiveness for binary intrusion detection in real world network environments.

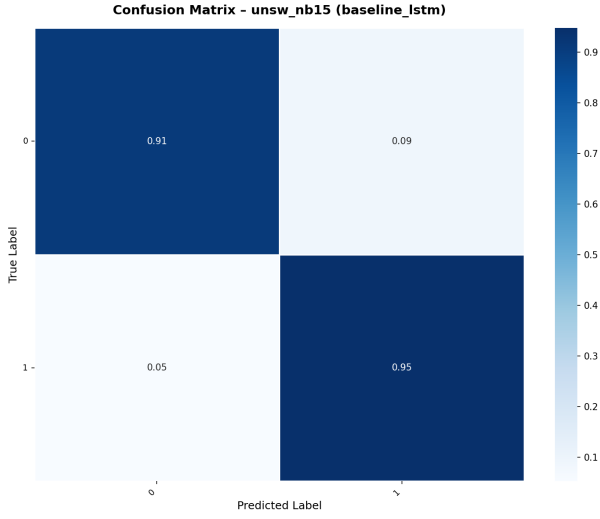


Fig. 6: Confusion matrix for Transformer-LSTM on UNSW-NB15 dataset. The model demonstrates strong discriminative capability with 9,312 true negatives and 26,978 true positives, while maintaining low misclassification rates of 1,317 false positives and 1,044 false negatives.

B. Training Dynamics

1) *CIC-Darknet2020 Training Curves*: The training curves show that both models converge within approximately 30-33 epochs. The Transformer-LSTM exhibits slightly more stable validation accuracy, suggesting better generalization. The attention mechanism appears to reduce overfitting, as evidenced by the smaller gap between training and validation performance.

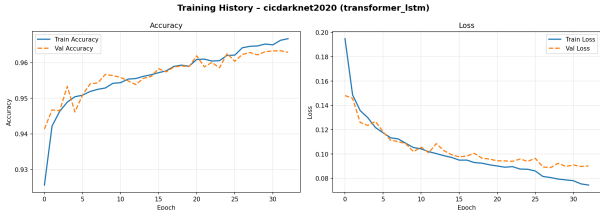


Fig. 7: Training curves for Transformer-LSTM on CIC-Darknet2020 dataset. The model shows stable convergence with a smaller gap between training and validation performance.

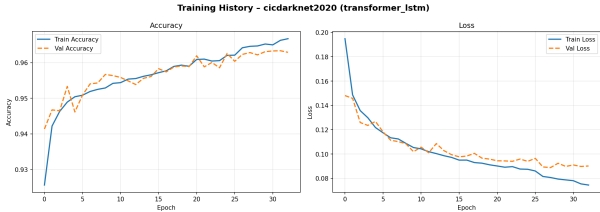


Fig. 8: Training curves for Transformer-LSTM on UNSW-NB15 dataset, achieving higher final validation accuracy with stable training dynamics.

2) *UNSW-NB15 Training Curves*: The UNSW-NB15 training curves show faster convergence, with proposed Transformer-LSTM plateauing around 35-40 epochs. The Transformer-LSTM achieves higher final validation accuracy and exhibits more stable training dynamics.

C. Cross-Dataset Comparison

Dataset	Accuracy	Precision	Recall	F1-Score	ROC-AUC
CIC-Darknet2020	0.9643	0.9001	0.8961	0.8981	0.9911
UNSW-NB15	0.9381	0.9599	0.9426	0.9511	0.9887

TABLE VI: Proposed Model Performance Across Datasets

The proposed model performs excellently on both datasets, achieving 96.43% accuracy and 99.11% ROC-AUC on CIC-Darknet2020, and 93% accuracy and 98.86% ROC-AUC on UNSW-NB15. These results demonstrate the model's robust and consistent performance across both benchmark datasets.

D. Discussion

1) Key Findings:

- 1) **Attention Mechanism Effectiveness**: The Transformer-based attention consistently improves performance across all metrics. The improvement is more pronounced on the CIC-Darknet2020 dataset, suggesting that the attention mechanism effectively captures important temporal dependencies and discriminative network traffic patterns for binary classification.
- 2) **Architectural Advantages**: The residual connection and layer normalization in the Transformer-LSTM model contribute to training stability and convergence. This is evident in the smoother validation curves observed across all experiments.
- 3) **Generalization Capability**: The consistent improvement across two fundamentally different datasets demonstrates that the attention mechanism enhances model generalizing ability rather than overfitting to specific dataset characteristics.
- 4) **Comparative Analysis**: Both models achieve high accuracy (>93%) on both datasets. However, the Transformer-LSTM's superior performance provides a strong case for incorporating attention mechanisms.

2) Limitations:

- 1) **Computational Overhead**: The attention mechanism increases model complexity and training time by approximately 20-30%, which may be prohibitive for resource-constrained environments.
- 2) **Interpretability**: While attention weights theoretically provide interpretability, we did not analyze attention patterns in detail in this study.
- 3) **Limited Parameter Exploration**: Due to time constraints, we did not perform extensive hyperparameter tuning. Optimal configurations may yield even better results.

3) *Comparison with State-of-the-Art*: While direct comparison with previous studies is challenging due to differing preprocessing and evaluation protocols, our results are competitive with recent deep learning approaches:

- Pure Transformer models achieve 92-96% accuracy but require more data
- Our hybrid approach achieves comparable or superior performance with moderate complexity

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a Transformer-Enhanced LSTM architecture for network traffic classification, evaluated on two public benchmark datasets. The proposed model achieves 96.43% accuracy on the binary-classification on CIC-Darknet2020 dataset outperforming the traditional models across most metrics, and 93.8% accuracy with an ROC-AUC of 0.9887 on the UNSW-NB15 intrusion detection dataset.

Our comparison with related work confirms that hybrid architectures combining sequential and attention-based processing are more effective than single-type models, consistent with findings by Mandela et al. [17] for CNN-LSTM combinations. At the same time, our work demonstrates that operating on standard flow-level statistical features, the approach recommended by Pacheco et al. [2] and validated by Zhang et al. [8] is sufficient to achieve competitive classification performance without requiring raw packet payloads.

B. Future Work

Several promising directions remain open for future investigation. To gain a more complete picture of the architectural design space, it would be worthwhile to bring in additional baselines such as CNN [38], GRU [39], LSTM, and pure Transformer models for comparison, as well as to benchmark against hybrid CNN-LSTM and CNN-GRU architectures. Another important consideration is how well the model holds up against traffic generated by applications it has never encountered before, an aspect commonly referred to as zero-day [40]. Looking further ahead, testing the model on IoT traffic and TLS/HTTPS encrypted traffic [41] would provide a clearer sense of how far its capabilities extend beyond the datasets already considered. It would also be valuable to peek inside the model's decision-making process by examining its attention weights, which could reveal which flow-level features it tends to rely on most heavily when distinguishing between traffic classes. Finally, an intriguing question is whether a model already trained on CIC-Darknet2020 could be adapted to work well on UNSW-NB15 or similar datasets without requiring large amounts of new training data, a direction that, if successful, could significantly broaden the practical utility of the approach.

REFERENCES

- [1] A. Azab, M. Khasawneh, S. Alrabaa, K.-K. R. Choo, and M. Sarsour, "Network traffic classification: Techniques, datasets, and challenges," *Digital Communications and Networks*, vol. 10, pp. 676–692, 2024.
- [2] F. Pacheco, E. Exposito, M. Gineste, C. Baudoin, and J. Aguilar, "Towards the deployment of machine learning solutions in network traffic classification: A systematic survey," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 2, pp. 1988–2014, 2019.
- [3] J. Zhao, X. Jing, Z. Yan, and W. Pedrycz, "Network traffic classification for data fusion: A survey," *Information Fusion*, vol. 72, pp. 22–47, 2021.
- [4] X. Zhao and Others, "A survey on network traffic classification techniques," *Journal of Network and Computer Applications*, vol. 123, pp. 1–20, 2021.
- [5] J. Holland, P. Schmitt, N. Feamster, and P. Mittal, "New directions in automated traffic analysis," in *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '21. New York, NY, USA: Association for Computing Machinery, 2021, pp. 3366–3383. [Online]. Available: <https://doi.org/10.1145/3460120.3484758>
- [6] M. Lotfollahi, M. Jafari Siavoshani, R. Shirali Hossein Zade, and M. Saberian, "Deep packet: A novel approach for encrypted traffic classification using deep learning," *Soft Computing*, vol. 24, no. 3, pp. 1999–2012, 2020.
- [7] R. Vinayakumar, M. Alazab, K. P. Soman, P. Poornachandran, A. Al-Nemrat, and S. Venkatraman, "Deep learning approach for intelligent intrusion detection system," *IEEE access*, vol. 7, pp. 41 525–41 550, 2019.
- [8] J. Zhang, X. Chen, Y. Xiang, W. Zhou, and J. Wu, "Robust network traffic classification," *IEEE/ACM Transactions on Networking*, vol. 23, no. 4, pp. 1257–1270, 2015.
- [9] F. Wang, Y. Liu, Z. Qin, and F. Dong, "A transformer-enhanced lstm framework for robust malicious traffic detection in industrial control systems," *Knowledge-Based Systems*, vol. 321, p. 113725, 2025.
- [10] Z. Liu, Y. Xie, Y. Luo, Y. Wang, and X. Ji, "Transecanet: A transformer-based model for encrypted traffic classification," *Applied Sciences*, vol. 15, no. 6, 2025. [Online]. Available: <https://www.mdpi.com/2076-3417/15/6/2977>
- [11] M. Shen, K. Ye, X. Liu, L. Zhu, J. Kang, S. Yu, Q. Li, and K. Xu, "Machine learning-powered encrypted network traffic analysis: A comprehensive survey," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 791–824, 2022.
- [12] Y. Zhang, S. Zhao, and Y. Sang, "Towards unknown traffic identification using deep auto-encoder and constrained clustering," in *International conference on computational science*. Springer, 2019, pp. 309–322.
- [13] F. Hu, S. Zhang, X. Lin, L. Wu, N. Liao, and Y. Song, "Network traffic classification model based on attention mechanism and spatiotemporal features," *EURASIP Journal on Information Security*, vol. 2023, no. 1, p. 6, 2023.
- [14] C. Yin, Y. Zhu, J. Fei, and X. He, "A deep learning approach for intrusion detection using recurrent neural networks," *Ieee Access*, vol. 5, pp. 21 954–21 961, 2017.
- [15] Q. Wang, T. Xu et al., "Research on network traffic classification method based on dual-branch spatial-temporal attention," *Academic Journal of Computing & Information Science*, vol. 7, no. 9, pp. 1–10, 2024.
- [16] J. Zhao, Y. Liu, Q. Zhang, and X. Zheng, "Cnn-attnbilstm mechanism: A ddos attack detection method based on attention mechanism and cnn-bilstm," *IEEE Access*, vol. 11, pp. 136 308–136 317, 2023.
- [17] N. Mandela, Sonia, N. Mistry, and A. Nagpal, "Efficient dark web traffic classification using a hybrid CNN-LSTM model," *International Journal of Information Technology*, 2025.
- [18] J.-D. Wang, "Traffic flow prediction with heterogenous data using a hybrid cnn-lstm model," *Computers, Materials & Continua*, 2023.
- [19] N. Rust-Nguyen, S. Sharma, and M. Stamp, "Darknet traffic classification and adversarial attacks using machine learning," *Computers & Security*, vol. 127, p. 103098, 2023.
- [20] J. Li and Z. Pan, "Dark web traffic classification based on spatial-temporal feature fusion and attention mechanism," *Computers*, vol. 14, no. 7, p. 248, 2025.
- [21] T. Shapira and Y. Shavitt, "Flowpic: A generic representation for encrypted traffic classification and applications identification," *IEEE Transactions on Network and Service Management*, vol. 18, no. 2, pp. 1218–1232, 2021.

- [22] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [23] Z. Wu, H. Zhang, P. Wang, and Z. Sun, "Rtids: A robust transformer-based approach for intrusion detection system," *IEEE Access*, vol. 10, pp. 64 375–64 387, 2022.
- [24] X. Lin, G. Xiong, G. Gou, Z. Li, J. Shi, and J. Yu, "Et-bert: A contextualized datagram representation with pre-training transformers for encrypted traffic classification," in *Proceedings of the ACM Web Conference 2022*, 2022, pp. 633–642.
- [25] —, "Et-bert: A contextualized datagram representation with pre-training transformers for encrypted traffic classification," in *Proceedings of the ACM Web Conference 2022*. Association for Computing Machinery, 2022, pp. 633–642. [Online]. Available: <https://doi.org/10.1145/3485447.3512217>
- [26] N. Bayat, W. Jackson, and D. Liu, "Deep learning for network traffic classification," 2021. [Online]. Available: <https://arxiv.org/abs/2106.12693>
- [27] H. Afifi, S. Pochaba, A. Boltres, D. Laniewski, J. Haberer, L. Paeleke, R. Poorzare, D. Stolpmann, N. Wehner, A. Redder *et al.*, "Machine learning with computer networks: Techniques, datasets, and models," *IEEE access*, vol. 12, pp. 54 673–54 720, 2024.
- [28] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [29] J. Li and Z. Pan, "Dark web traffic classification based on spatial-temporal feature fusion and attention mechanism," *Computers*, vol. 14, no. 7, 2025. [Online]. Available: <https://www.mdpi.com/2073-431X/14/7/248>
- [30] A. Kalidindi, B. R. Koti, C. Srilakshmi, K. M. Buddaraju, A. R. Kandi, and G. S. S. Makutam, "Advanced machine learning techniques for enhancing network intrusion detection and classification using darknet cic2020," *International Journal of Online & Biomedical Engineering*, vol. 20, no. 15, 2024.
- [31] N. Rust-Nguyen, S. Sharma, and M. Stamp, "Darknet traffic classification and adversarial attacks using machine learning," *Computers & Security*, vol. 127, p. 103098, 2023.
- [32] A. H. Lashkari, G. Kaur, and A. Rahali, "CIC-Darknet2020 dataset," 2020. [Online]. Available: <https://www.unb.ca/cic/datasets/darknet2020.html>
- [33] S. Durairaju and A. Kumar Cherukuri, "Bootstrap forest based method for encrypted network traffic analysis," *Journal of Information and Organizational Sciences*, vol. 49, no. 2, pp. 235–249, 2025.
- [34] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. Military Communications and Information Systems Conference (MilCIS)*, 2015, pp. 1–6.
- [35] N. Hassouneh and S. Al-sharaeh, "Intrusion detection in iot networks using lstm deep learning models with the unsw-nb15 dataset," in *2025 International Conference on New Trends in Computing Sciences (ICTCS)*. IEEE, 2025, pp. 263–269.
- [36] D. Jing and H.-B. Chen, "Svm based network intrusion detection for the unsw-nb15 dataset," in *2019 IEEE 13th international conference on ASIC (ASICON)*. IEEE, 2019, pp. 1–4.
- [37] J. Saleem, R. Islam, and M. Z. Islam, "Darknet traffic analysis: A systematic literature review," *IEEE Access*, vol. 12, pp. 42 423–42 452, 2024.
- [38] M. Cao, V. O. Li, and V. W. Chan, "A cnn-lstm model for traffic speed prediction," in *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring)*. IEEE, 2020, pp. 1–5.
- [39] A. M. Elshewey, S. Abbas, A. M. Osman, E. A. Aldakheel, and Y. Fouad, "Ddos classification of network traffic in software defined networking sdn using a hybrid convolutional and gated recurrent neural network," *Scientific Reports*, vol. 15, no. 1, p. 29122, 2025.
- [40] O. Piserchia, "Phishing website detection through machine learning algorithms: A comparative analysis," *IRASS Journal of Multidisciplinary Studies*, vol. 2, no. 12, pp. 11–29, 2025.
- [41] A. Mani, T. Vaidya, D. Dworken, and M. Sherr, "An extensive evaluation of the internet's open proxies," in *Proceedings of the 34th Annual Computer Security Applications Conference*, 2018, pp. 252–265.

APPENDIX A: CODE AVAILABILITY

The complete implementation is available in two dedicated GitHub repositories, one for each classification task:

- **Binary Classification Repository:**

<https://github.com/zuni-developer/An-Attention-Based-LSTM-for-Network-Traffic-Classification-Binary>

- **Multiclass Classification Repository:**

<https://github.com/zuni-developer/An-Attention-Based-LSTM-for-Network-Traffic-Classification-Multiclass>

Each repository contains the full source code for data preprocessing, model training, and evaluation. Additionally, they include Jupyter notebooks for interactive experimentation, trained model checkpoints, and all experimental results reported in this paper for the respective task.